



Sell Your Power. Profitably.

Steamaco IoT-AMI Smart Metering Platform

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Version: V2 R1



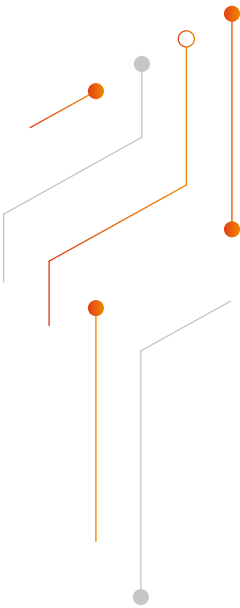
Steamaco Company Limited

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imagine if selling
energy was easy...



What if you could understand each consumer in detail, collect their payments with ease, eliminate your commercial losses and deliver an efficient and reliable energy service?

What impact would it have on your revenue and profit?

Right now, we know this isn't the case.

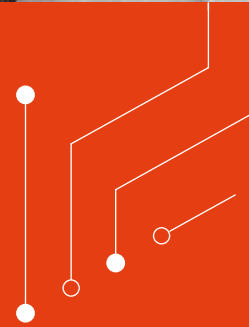
A lack of visibility of consumption is making it impossible for you to collect the revenue that's rightfully yours.

And with more than 95% of utilities in Africa making a loss, the pressure is on to find a reliable and effective solution.



02. your smart metering solution

operate more effectively.
detect and resolve losses.
build consumer trust.



SteamaCo's SAMI platform is the answer.

Our Advanced Metering Infrastructure solutions give you the ability to sell power profitably and easily by overcoming your most high-value challenges:

- 1. Collecting revenue**
- 2. Detecting and resolving ATC&C losses**
- 3. Engaging consumers**
- 4. Automating daily power distribution operations**

You'll have the data and tools to improve revenue assurance, reduce operational costs and build consumer trust.

That means more stability for the company and your employees, more confidence amongst your senior leadership and less pressure on you and your team.

We'll work with you to understand your needs and tailor a smart metering solution. We'll help you implement and integrate it into your business and provide daily support to make sure you maximise the potential of the platform.

By partnering with SteamaCo, you can deliver reliable energy to every one of your consumers, in both urban and rural areas. Plus, contribute to a better future for energy consumption in Africa, Asia and beyond.

Together, we can create a better future.

introducing steamaco

SteamaCo offers a scalable IoT smart metering (IoT-AMI) platform and turn-key solution for managing electricity sales more effectively.

Our cloud-based platform helps you get reliable, accurate data on usage and theft, so you know who's used power and where to intervene to resolve a loss of revenue.

With SteamaCo, you can:

Enhance your revenues

Without visibility of consumption, you're managing your distribution grids in the dark. The SteamaCo platform turns on the lights, allowing you to see who's using your electricity, where, when and how much.

Gain revenue assurance

Manually searching for theft incidents is time-consuming, impractical and ineffective. Using AI to detect anomalous usage and pinpoint instances of theft, our apps enable you to resolve these instances effectively in the field.

Reduce operational costs

Labour-intensive processes drain your ability to invest and progress. Through our operational automation and alerts, you can lower expenses and achieve better asset control.

Integrate with existing systems

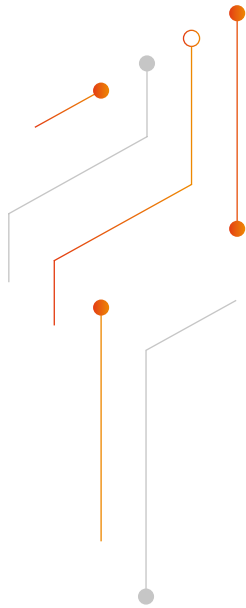
Building workarounds to let legacy systems interact with billing and metering solutions is impractical, expensive and imperfect. The SteamaCo platform is a frictionless API-led solution, fully-interoperable and easy to plug and play.

Build consumer trust

If your customer communication is uncoordinated and inconsistent, it could erode confidence in your service. The platform enables you to achieve multi-channel, real-time customer engagement.



the interoperable solution



The SteamaCo platform is fully integrated and interoperable with a range of hardware and enterprise systems, including:

- *IEC standard DLMS/COSEM compliant metering equipment;*
- *Third-party Head End Systems (HES) and Meter Data Management Systems (MDMS);*
- *Prepaid energy vending systems and payment systems (STS vending platforms and mobile money providers);*
- *Post-paid billing and payment systems, generation and distribution management systems;*
- *Other enterprise systems, including ERP and CMS systems, and many more.*

It provides you with:



Secure data sharing

between all your systems
via a RESTful Application
Programming Interface (API)



Full integration

with any other Head End Systems
and/or Meter Data Management
systems that you operate



Seamless configuration

with your wider operational
policies and regulated
operational processes



Quick customisation

to new requirements with fast
implementation, testing and
deployment by our engineers



capabilities & benefits

The SteamaCo platform is a secure, scalable and reliable Software-as-a-Service (“SaaS”) platform, with a W3C-compliant web-browser interface.

You can access the platform anywhere, anytime.

PLATFORM CAPABILITY	BENEFIT
1. Automatically acquire meter data, frequently. Meter readings, alarms and power quality metrics are received at a pre-defined schedule or on-demand. Automatically re-read meters to acquire missing data.	You get up-to-date consumer and operational intelligence as often as you need but without the cost of reading meters manually. Never lose meter data and operate a smarter grid.
2. Remotely control smart meters. Remotely switch meters off or on to disconnect or reconnect the energy supply to each consumer.	You have full control over how and when you supply energy to your consumers. React quickly in emergency situations without needing to go to the meter.
3. Automate daily utility operations. Apply a variety of tariffs, time-of-use (ToU) policies and schedules, power limits and free power allocations, as needed, to deliver a fully automated and tailored energy supply to your consumers.	Increase profitability and reduce the cost-to-serve by operating the distribution of energy highly efficiently, reliably and automatically at much lower daily operating cost.
4. Remotely manage and configure all of your smart metering devices. Use the platform to keep a record of, and update, all of your equipment configurations across your distribution network.	Reduce the costs, time and complexity of manually managing and reconfiguring your distribution network’s smart devices. Plus, configuration changes are automatically synchronised to the equipment.

PLATFORM CAPABILITY

BENEFIT

5. Apply firmware updates 'Over-the-Air' (OTA). For devices that support OTA, firmware can be remotely upgraded through the platform. Firmware upgrades can add new functionality to the device over its physical lifetime.

Future-proof your investment in smart metering hardware and keep O&M costs low. The platform lets you update OTA devices quickly and easily.

Extend the useful lifetime of your smart meters and avoid costly, complex device upgrade or replacement projects.

6. Transmit STS prepayment / credit tokens to the meters electronically with remote crediting. Consumers can buy online or at an agent or via mobile money, and have the credit applied automatically and immediately.

Make power easy to buy, provide a positive consumer experience and build trust in your brand. You can reliably keep pay-to-power time to minutes.

Consumers can buy power wherever they are, even when away from their home or business, or on behalf of others.

7. Accept other forms of payments. Cash (via Agents) and Mobile Money payments (via platform integration) and apply the credits to consumer account balances and to the meters.

As above. Plus, you can reduce unnecessary costs by not requiring an STS vending platform to be operated and maintained alongside the AMI.

8. Record all payment and usage transactions in an indelible ledger. The ledger keeps record of customer payment data from STS and mobile money vending systems and agents. Plus it keeps record of all usage charges billed to the consumer from each meter reading.

Have detailed, accurate and secure transaction data available for reporting.

Gain detailed consumer intelligence due to marrying vending data (energy sales data) with usage data (meter readings) using SteamaCo's analytics functions.

9. Communicate with consumers via SMS or the Consumer Site.

Send messages to groups and individuals, warn them of low balances and keep them informed about important service announcements. Confirm payments when they are received. Consumers can access their full account, usage, payment and tariff data in the Consumer Site.

Provide a positive consumer experience and reduce the demand and costs for customer support services.

Plus, build trust among consumers by making it easy for them to check their balances at any time using their mobile or smart device from anywhere. Consumers stay fully informed of their energy consumption using the Consumer Site.

10. Provide comprehensive charts and usage reports down to individual meters.

The Chart Builder tool allows you to quickly create detailed charts that display data as you require. Share these charts with other operators. Every site, meter and consumer account page displays the last 7 days of energy data in a chart.

Understand your operations easily with visual data.

Build trust with consumers by providing a positive consumer experience, and making it easy to deliver excellent customer support and respond when consumers have a query.

11. Display power and diagnostic metrics visually on site maps.

Quickly see where common faults or supply issues are occurring and what kind of fault. Understand the issue that is occurring before you head out to the site.

Improve the overall quality and reliability of the energy supply service across the grid, while reducing the costs of maintaining it — especially for distant/inaccessible sites — and minimising the need for site visits to troubleshoot and resolve faults.



01. collecting revenue

Payments and pre-paid meter switching

The platform gives you complete flexibility to collect your revenue, whether you operate pre-paid and post-paid smart meters.

For pre-paid meters, you don't need to use an STS vending platform to accept payments. Nor do consumers need to input STS credit tokens into their meters to switch their power on.

Consumers can pay for their electricity usage in two ways without using STS tokens:

*Mobile money via their mobile phone; and
Cash or other accepted method via utility agent*

The platform is capable of remotely, automatically switching consumer pre-paid meters off or on based on when their credit balance reaches zero credit, while their pre-paid top-ups add credit to their balance.

However, if you prefer to use STS to sell power and add credit directly to meters, the platform can be integrated with your STS vending platform. It can send STS tokens to the consumers' smart meters automatically. Or you can still allow your consumers to input tokens manually if you prefer.

Once credited, the meter manages switching energy supply autonomously, while the platform monitors each consumer's usage and credit balances.

All payment notifications are recorded within the platform's indelible ledger. The platform can send an SMS to the consumer confirming that their payment has been received.

You can also encourage customers to purchase energy in larger amounts by incentivising them with a bonus credit (e.g. receive 10% more credit by topping up above a threshold amount).

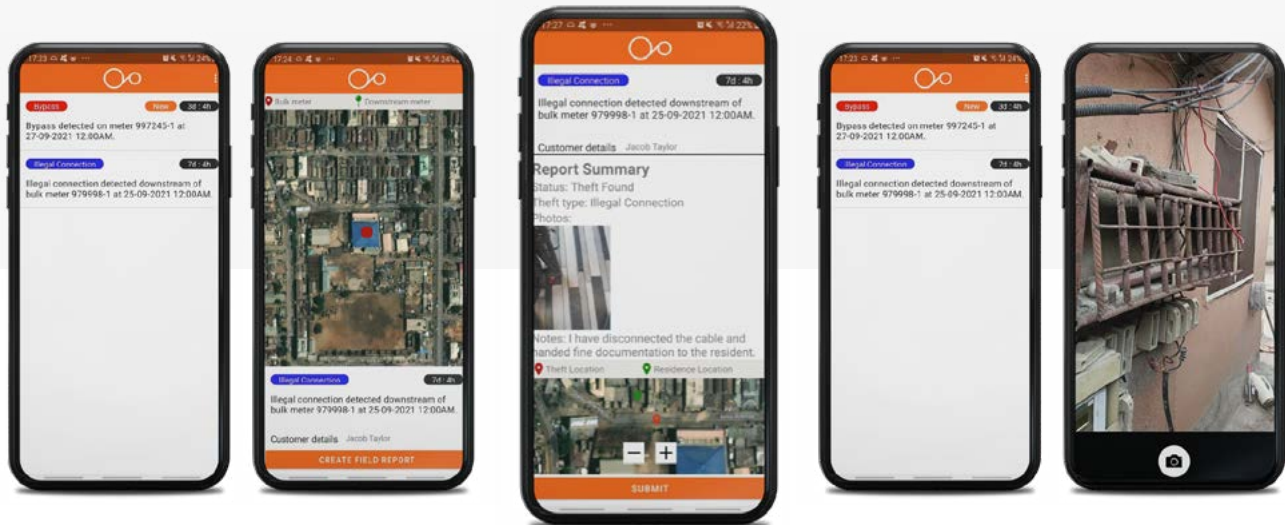
This works in a similar fashion to airtime and data bundles offered by many pay-as-you-go telecom companies.

Energy usage charging

The platform automatically takes a reading from the meter once every hour (although you can change this as you need) and applies a charge to the consumer's account.

The charge is calculated according to their set tariff and the transaction amount and date/time is recorded in the platform's indelible ledger.

02. detecting & resolving ATC&C losses



ATC&C Loss Detection & Resolution

The IoT-AMI platform dramatically improves your profitability from supplying power using simple customer information, smart meter data and hunting for anomalies.

By gathering available data from across the distribution network's metering equipment (both consumer meters, plus transformer meters and supply-line totalizer meters) and from your payment vending system, the platform detects anomalies in consumption and payments that indicate when excessive ATC&C losses occur.

Alerts are generated in the platform to notify you and losses can be visualised in a chart for each distribution line. You can then perform an investigation and carry out corrective actions.

Continual innovation to assure revenue

Theft of power is a significant problem and is a significant drain on revenues for utilities of all sizes across Africa.

SteamaCo is hard at work adding innovative new tools to the platform that will deliver you even greater control over detecting and resolving losses caused by theft and fraud.

03. engaging consumers

Consumer notifications

Automated notifications sent by the platform keep consumers informed and in control of their usage, and drive revenue collection for you.

An informed consumer is also less likely to complain to you if they understand their usage and are warned before their supply disconnects.

Consumers receive notifications by SMS or via the Consumer WebApp:

- *When their prepaid credit balance is low, to encourage them to make a top-up payment;*
- *When they run out of pre-paid credit and their meter is switched off, so that they know to buy additional credit;*
- *When a post-pay consumer's usage reaches a certain threshold. This warning empowers them to avoid an unexpectedly large bill;*
- *When payments are received and supply switched on again, to give peace of mind.*

A record is kept of all consumer notifications in the platform, allowing the cost benefit of your level of consumer engagement to be assessed.

Consumer WebApp

The Consumer WebApp allows consumers with internet access to view their account information held in the platform, check their credit balance, transaction history, and energy usage data (in Currency & kWh) wherever and whenever they want to. It provides consumers with detailed information on payments and power usage, building trust with you.

Optional CIU devices or LCD displays

The Consumer WebApp can display all of the information required to be displayed via the meter's display or customer interface unit (CIU) to a customer. This means that customers who can use the WebApp, don't need to have a CIU device installed with their meter.

Ease of use

Customers with WebApp access can check their account & energy usage information at any time and wherever they have access to the internet.

Secure

Customers require a username and password to access their account data via the WebApp.

Included in the WebApp are:



Meter status



Usage history
(daily & monthly)



Spend history
(daily & monthly)



Transaction history



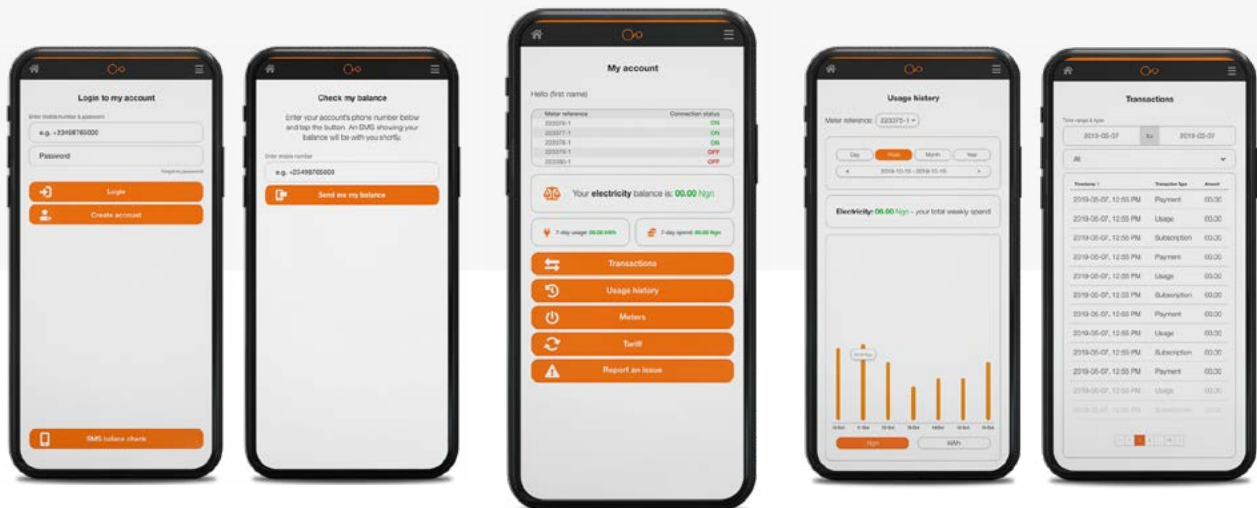
Tariff details



Issue reporting
(coming soon)



Embedded FAQs
(coming soon)



Illustrative screenshots from our Consumer WebApp

Crystal clear information

The Consumer WebApp uses charts and graphs to make it easy for a consumer to understand their energy usage and payments.

This data is also presented in a chart for each consumer account within the Web UI, which makes it easier for your customer support team to help consumers when they have a query about their account or usage.

Consumer SMS messages can be sent in the consumer's language and are designed to be easy to understand.

04. automating daily power distribution operations



Power distribution operations require perfectly coordinated, repetitive execution. Critical tasks like acquiring meter data, amending consumer balances and switching power are prone to human error when performed manually, and the problem gets worse with scale.

The platform automates your daily operations in real time. You determine the operating policies, the platform then handles the execution of those policies, moment by moment, automatically.

The platform lets you automate:

- *Consumer meter switching*
- *Meter data acquisition*
- *Consumer notifications*
- *Usage charging according to the set tariffs*
- *Time of use power limits*
- *Scheduled and emergency outages*
- *Alerts when issues arise*

The platform's Web UI provides you all the tools you need to configure, monitor and report on your power distribution operations.

Clean, Simple and Effective UI

The User Interface ('UI') for authorised system managers and administrators is provided through a web browser on any computer or mobile device with internet access.

Access to the platform's Web UI is protected by password authentication and access to user functions is granted by the system administrator.

The SteamaCo platform allows for individual user access rights (or permissions) to various features, reports, user information and configuration pages within the UI. The permissions are fully configurable for each user account.

The Web UI is fully backed by the REST API and all operations possible on the UI can be carried out through API calls. It allows you to configure settings and see the status of your operations:

Data visualisation tools

KPI dashboards show the performance of your entire portfolio

Charts showing key metrics for the last 7 days are shown for each Site and each Meter

Custom dashboards can be produced on-demand

Flexible Chart Builder allows you to visualise performance trends for various time-series metrics, at Utility, area, consumer, or meter level:

- *Build the specific charts you need and share with colleagues*
- *Export each chart's data into a CSV file*

A chart of recent usage and account balance is shown for each consumer account, makes supporting consumers easier

Remote asset management

Maintain up-to-date information on all your distribution assets

Perform automated and manual meter operations

Identify at a quick glance which meters need further investigation

View colour-coded site maps showing signal strength, uptime, voltage on an individual meter basis

Automated emails sharing details of ongoing communication issues

Consumer management

Single or bulk consumer registration and configuration

List view showing consumer details, meter status, account balance, and 30-day kWh usage

Highly customisable payment plans and tariffs on per-consumer basis

Consumer communication settings

Customise automated consumer SMS for platform operations such as line switching and payment notifications in your consumer's desired language

Perform ad-hoc individual or bulk consumer messaging

Use SMS shortcodes that are toll-free for the consumer.

National USSD-shortcode systems can be integrated with the platform via the API

Sites

Define multiple sites in the platform as needed. Each site is used to group hardware logically according to a defined location (an area, a community, a neighbourhood, a village, etc.) or a hardware cluster or private wire sub-network within your distribution network

Sites can be used to aggregate energy consumption and payment data to site-level totals

Apply settings (e.g. meter configurations, ToU policies, Tariffs) quickly, in bulk to all smart hardware at the site.

Tariffs & Credit Bundles

A wide variety of Tariffs are available to suit your needs:

- *Flat rates and per kWh rates*
- *Subscriptions and minimum top-ups*
- *Hybrid tariffs and monthly usage tiers*
- *Connection fee pay-off plans*
- *We'll configure your special tariff if you need*

Apply tariffs site-wide or to individual consumers

Configure credit bundles to incentivise larger value top-up purchases

Time of Use (ToU) policies

Configure policies to set variable power limits daily and apply site-wide or to individual consumers

Schedule regular or ad-hoc outages (e.g. daily shutdown 1am - 5am)

Prevent excessive usage site-wide by allowing different consumer groups to use power intensive appliances at different times of day

Alerts and System Indicators

Automated Alerts from meter data and system analytics to pinpoint and investigate issues quickly

System indicators highlight meters that may need troubleshooting or investigation

Reports

Various customisable meter uptime and consumer usage CSV reports

Custom reports can be produced on demand

Audit logs

Track all changes to system entities by each user in a log, with a date and time stamp against each log entry

Tags

Create Tags and use them to characterise and group data records, making it easier to call for specific data through the UI and API



Illustrative screenshots from our Web UI application



05. interoperability & scalability

Full Data Transparency

The SteamaCo platform is fully interoperable with the other systems that you use, delivering complete data transparency.

It includes a web-based, browsable, REST API and has clear, extensive documentation to allow developers to integrate with the platform quickly and easily.

The API provides your other systems with full access to your data held in the platform.

It also allows these systems to share data with the platform, from payment transactions, to customer information systems, to data from your generation assets that aid in automating daily power distribution operations.

Using the API, you can easily retrieve information, configure assets, trigger operations, and create new records.

The API allows all connected systems to stay automatically updated, aligned, and consolidated via the SteamaCo platform.

A unique Integration ID can be stored against each customer account to help with integration with ERP, billing and other third-party systems.

Scalability

The platform is a fully distributed, cloud-based system that allows rapid scaling up on-demand additional servers and computational power.

The platform's segregated queuing systems and parallel processing means that, if there is a delay in processing one set of data due to outages or comms issues from a hardware grouping, it won't affect other data sets' processing times.

06. communications management

Smart Meter Communications and Data Efficiency

Data efficiency is key to keeping your operating expenses for smart meter communications low, while allowing the platform to acquire data frequently enough, and detailed enough, to deliver the operational capabilities of your distribution networks that you desire.

The platform helps you operate reliably and cost-effectively, even when GSM connectivity is scarce or disrupted. Cutting-edge network architecture and robust connectivity management technologies offer reliable operations even in low-connectivity environments or when connection is interrupted.

During commissioning, our support team works with you to ensure the smart meters are connected and configured correctly. This ensures they will provide only the data points required, at the intervals required, balanced to meet your operational needs and data cost requirements.

Head End System (HES)

The platform acts as a Head End System (HES), helping you manage your smart meter assets across your distribution network. It manages communications, configuration profiles and data acquisition autonomously, for all connected smart metering hardware.

It manages data ingesting efficiently, ensuring that even delayed data is properly processed in the time order in which it was recorded by the metering equipment.

Smart Meter Connectivity Problem Diagnosis and Resolution

Connectivity from smart meters to management software, irrespective of communications pathway, can be a challenge due to external factors (such as network issues).

The platform detects connectivity issues when they occur, sends an alarm to your operational teams and, through automation, remotely seeks to rectify the issue by looking to re-connect to the network. This functionality gives transparency of the issue to operational teams and reduces operational costs due to remotely fixing the communication fault.

Smart meter data acquisition is only delayed by temporary connectivity problems — it's never lost. This means meters continue critical operations even when communications are down. Data can be backed up on meters or communication gateways for up to 90 days, syncing when the connection is restored.

07. system operations & security

Meter Data Management

The platform provides an advanced, cloud-based meter data management system (MDMS) that works seamlessly with all of its other capabilities.

The API interface allows you to easily integrate other HES, and to provide accurate, processed meter data to your other enterprise systems.

Uptime and Performance Monitoring

Cloud Service

Steamaco uses Amazon Web Services (AWS) as our cloud service provider for the platform, and we maintain service uptime to >99.999%.

The cloud servers are deployed in a multi-tiered environment with comprehensive backup, failover and security services provided by AWS.

Notifications

Advance notifications are provided for any scheduled downtimes for maintenance and/or implementing upgrades and/or improvements.

System performance

This is proactively monitored at all times for our customers. Automated alerts are generated for any events that may impact system operations.



Self-resolving issues

The platform is capable of self-resolving issues and minimising their impact on your operations. It is designed to automatically deal with a wide array of faults or interruptions automatically.

Communication outages

Any prolonged communication outages do not impact system performance, data integrity and the delivery of power to consumers who have paid for it.

Maintaining high uptime for the communications between meter hardware and the AMI is a key component of the platform, with multiple points of redundancy and data storage built-in.

In the event of communication outages that may impact any part of the distribution network equipment, all delivery of data between the equipment impacted by the outage and the platform is merely delayed, not lost.

The data is then processed immediately on the restoration of the communication link.

Security and Data Protection

Amazon Web Services is used to host and operate the IoT-AMI platform and ensures a secure, certified Information Security Management System (ISMS).



Multi-tiered server environment



99.9999% uptime



End-to-end data encryption



Secure data access and user authentication



Regional Data Storage and transmission control



EU Data Protection and GDPR compliant



Automatic digital security certificate management

The platform is secured against unauthorised use, misuse and unauthorised data access. For instance:

Permissions-based User Access Control

This prevents unauthorised access to your system. Individual users can only view or edit what they are allowed to.

Audit logs

This keeps a record of every user's actions performed on the system.

Data privacy

Each customer can only access their data via their UI logins or their API keys. AWS cloud services are certified compliant with key standards, including:

- *ISO/IEC 27001:2013, 27017:2015, 27018:2019, 27701:2019, 9001:2015,*
- *CSA STAR CCM v3.0.1*

08. support services for customers



Steamaco believes in providing unrivalled support to each one of our customers. Our goal is to get you operational quickly and to support you to deliver access to energy, profitably, at all times.

We help you deploy quickly and efficiently

We provide you with all the training and support you need to deploy each meter at each site quickly and efficiently. You get your operations started sooner and deliver power and earn revenue faster.

Site mapping and deployment planning

We help and advise you on how to plan the deployment of your smart meter equipment correctly and ensure each device is capable of communicating with the platform.

Initial onboarding

We provide you with full training to familiarise you with key features: how to configure and operate meters, set up sites, customers, tariffs, ToU schedules, SMS communications, and more. We show you how to use our support features: the online Help Centre and our Help Desk (to raise support tickets).

Equipment configuration and testing

We train you on how to configure and test meters in the field, to ensure the meters can operate as you require and communicate reliably with the platform.

Platform deployment and set-up

We support your teams on-site and in the operations centre, and can provide any further training as needed, during your deployment projects. We show you how to set up the platform and import data.

We assist you to set up the integrations you need, such as mobile money & STS vending systems, and other API linked software you need, e.g. Odyssey.

Commissioning support

We support you to troubleshoot any issues with meter connections and platform operations, as you brings each site to go-live readiness.

We ensure your teams know how to use the Help Desk and support ticketing system so that you get the help you need without delay.

Monitoring post go-live

We keep an eye on your operations and system performance to ensure that your site is operating optimally. We remain highly responsive to support requests.

We make sure you keep operating efficiently

Proactive alerts are provided for potential issues and our expert support team works with you to resolve problems fast.

Step-by-step guidance is available online in the digital Help Centre that is built into the UI. Our detailed knowledge base of articles is always available to any user.

The Support Desk is available within the UI for any issues that cannot be resolved by following the step-by-step guidance, allowing you to raise a support request ticket that will be resolved in line with predetermined SLAs. All tickets are tracked until the issue is resolved.

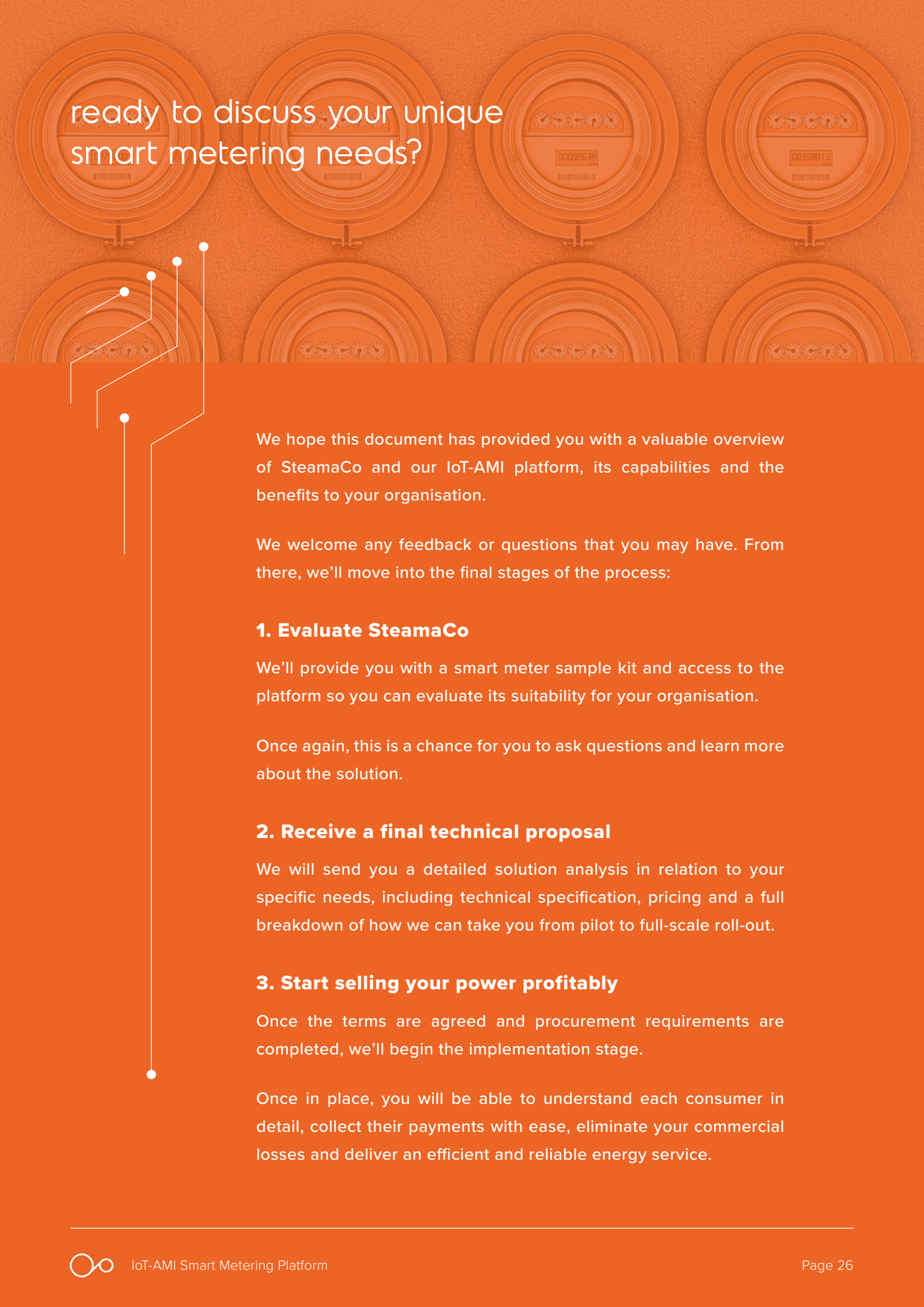
Feature development services

SteamaCo helps you deliver an energy service that evolves with your consumers. New features can be added to the platform to support evolving needs.

We co-create with you, ensuring that our product development is agile and responsive.

Smart today, Smarter tomorrow

With a one-time investment in metering hardware, continual improvements over the lifespan of the metering assets can be applied automatically, and over-the-air. Updates happen automatically, without requiring site visits or hardware replacement.



ready to discuss your unique
smart metering needs?

We hope this document has provided you with a valuable overview of SteamaCo and our IoT-AMI platform, its capabilities and the benefits to your organisation.

We welcome any feedback or questions that you may have. From there, we'll move into the final stages of the process:

1. Evaluate SteamaCo

We'll provide you with a smart meter sample kit and access to the platform so you can evaluate its suitability for your organisation.

Once again, this is a chance for you to ask questions and learn more about the solution.

2. Receive a final technical proposal

We will send you a detailed solution analysis in relation to your specific needs, including technical specification, pricing and a full breakdown of how we can take you from pilot to full-scale roll-out.

3. Start selling your power profitably

Once the terms are agreed and procurement requirements are completed, we'll begin the implementation stage.

Once in place, you will be able to understand each consumer in detail, collect their payments with ease, eliminate your commercial losses and deliver an efficient and reliable energy service.



ready to get started?

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